PROJECT REVIEW 2

# Employee Performance Evaluation System

A role-based Java web application for employee records, performance evaluations, goals, feedback and reports.


TEAM MEMBERS

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Galgotias University | B.Tech CSE (AI & ML)


**EPES**

PEOPLE • PERFORMANCE

**Java 25**

**Jakarta Servlet / JSP**

**MySQL 8 • Tomcat 11**

# 1. Project Overview

Why EPES was developed

## The challenge

Employee information, performance reviews, goals and feedback can become scattered across manual records and separate processes.

## The solution

EPES brings employee records and performance workflows into one web application with separate Manager and Employee access.

## PROJECT OBJECTIVES

- Centralize employee information and performance records.
- Support structured evaluations, ratings and manager feedback.
- Assign goals and track progress.
- Give employees secure access to their own performance information.

# 2. User Roles & Workflow

Two role-specific experiences share one database

## MANAGER

Sign in → Manage employees → Create evaluations → Assign goals → Provide feedback → View reports

## EMPLOYEE

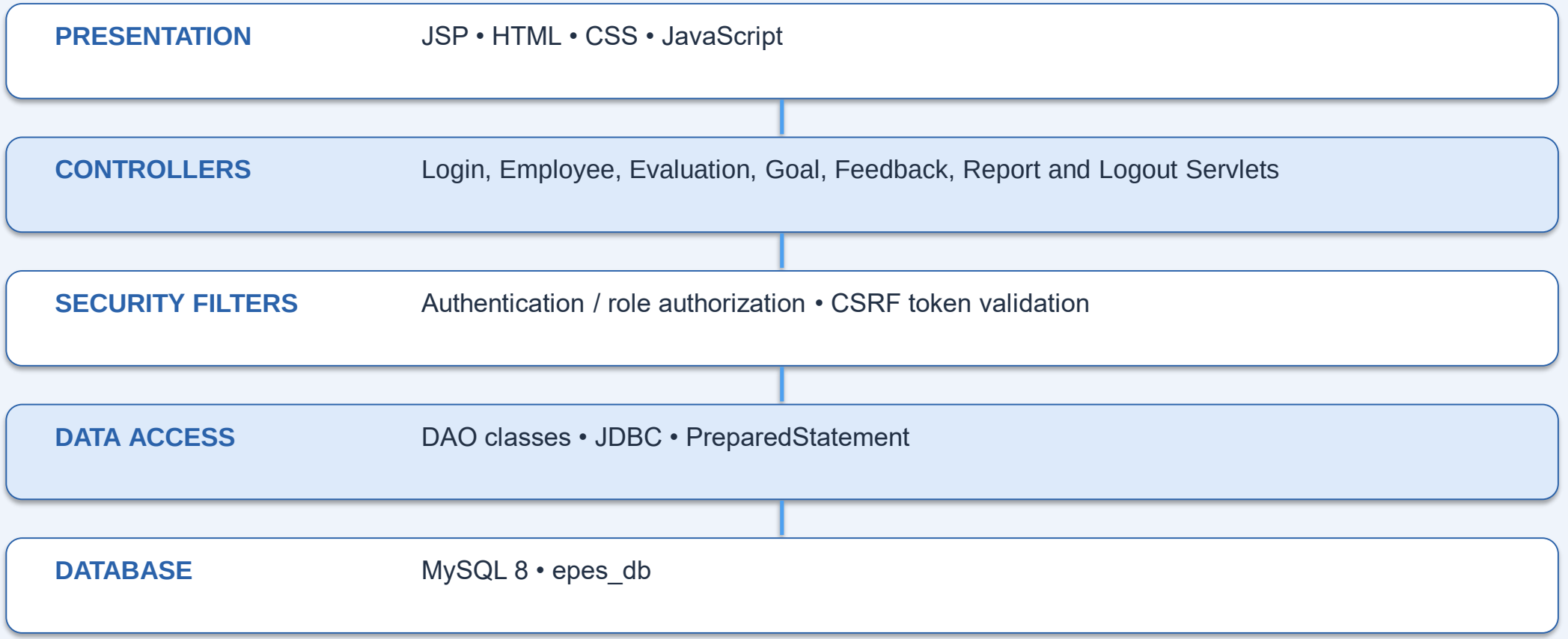
Sign in → View personal dashboard → Review evaluations → Track assigned goals → Read feedback and reports




Authorization and CSRF checks protect the request flow before data-changing operations are processed.

# 3. Application Architecture

Layered design separates the interface, request handling, security and data access



# 4. Functional Modules

Features available to Managers and Employees

## Employee Management

Add, view, update and delete employee records. Manager-only access.

## Performance Evaluations

Managers record evaluation criteria, ratings (1–5) and comments.

## Goals & Feedback

Managers assign goals and provide feedback; employees view their own records and update goal status.

## Performance Reports

Application calculates evaluation counts and average ratings for performance summaries.

# 5. Database Design

Eight MySQL tables support employee and performance workflows

### employees

PK employee_id

name • department • job_title

### users

PK user_id

email • password_hash • role

FK employee_id

### evaluations

PK evaluation_id

FK employee_id

criteria • rating • comments

### feedback

PK feedback_id

FK employee_id

rating • comments

### goals

PK goal_id

FK employee_id

description • target_date • status

### teams

PK team_id

team_name

### team_members

PK/FK employee_id

PK/FK team_id • role

### performance_reports

PK/FK employee_id

total_evaluations • average_rating

Foreign-key relationships connect performance records and user accounts to employees; team_members links employees and teams.

# 6. Security & Data Integrity

Security controls are applied across login, requests and database operations

## Password storage

PBKDF2WithHmacSHA256, per-password random salt.

## Role-based access

Manager and Employee permissions are checked on protected routes and actions.

## CSRF protection

Session token is rotated at login and validated for state-changing POST requests.

## Session handling

Fresh authenticated session; HttpOnly cookie; cookie-only session tracking.

## SQL safety

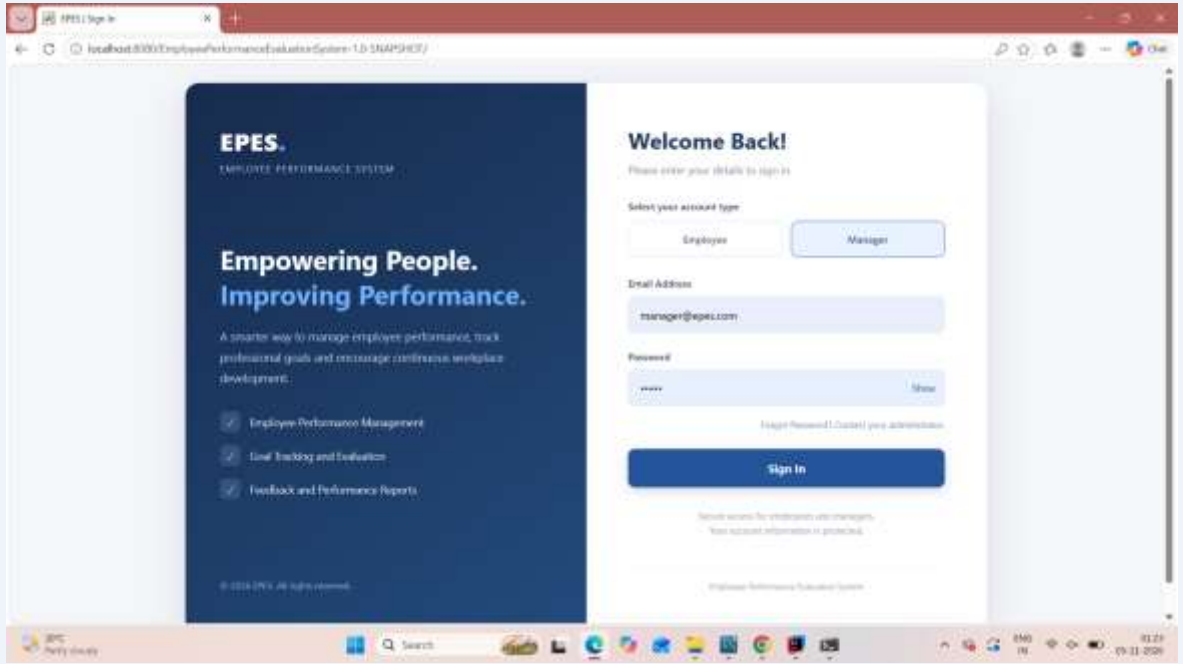
PreparedStatement parameters and try-with-resources for JDBC resources.

## Validation

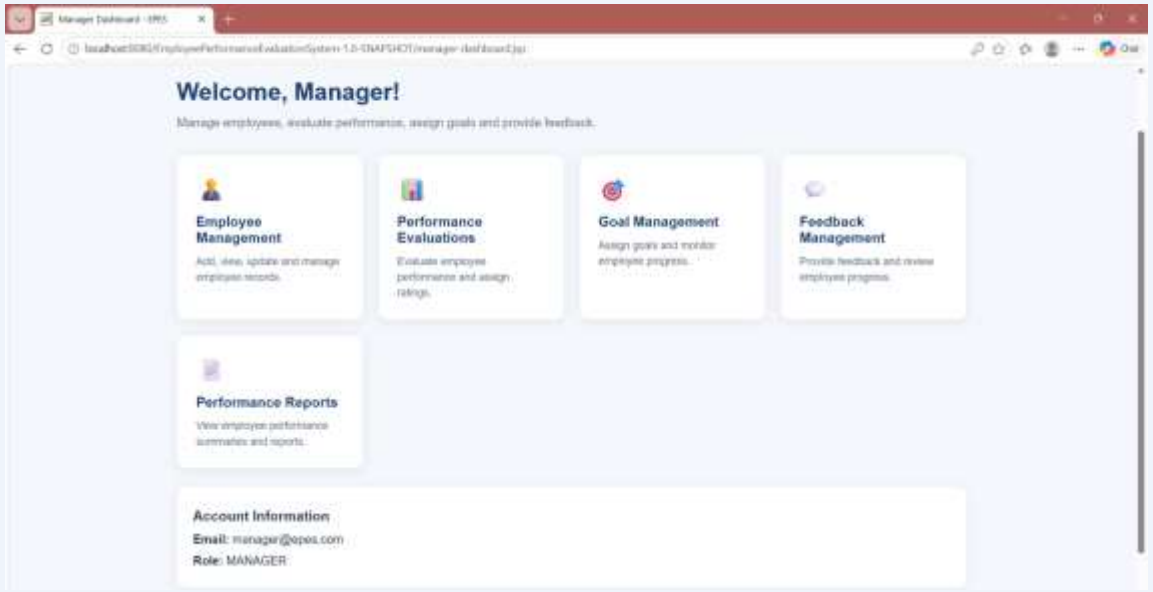
Server-side checks plus database CHECK constraints for ratings and goal status.

# 7. Application Screenshots — Login & Manager

Actual screenshots captured from the local application



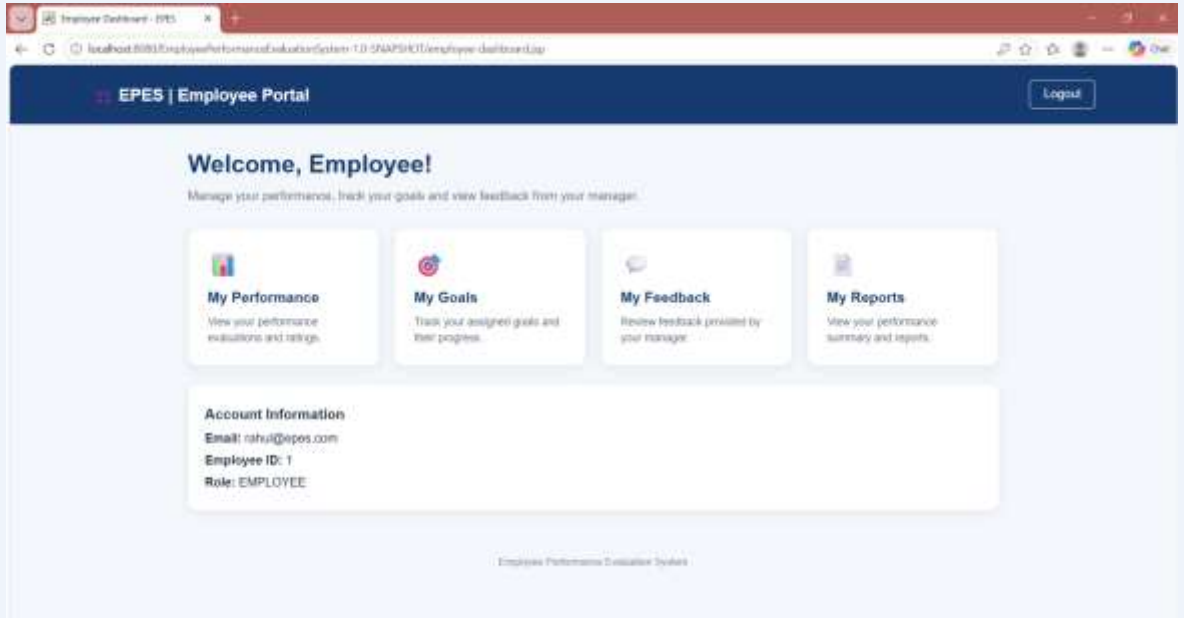

Login page with Employee / Manager selection



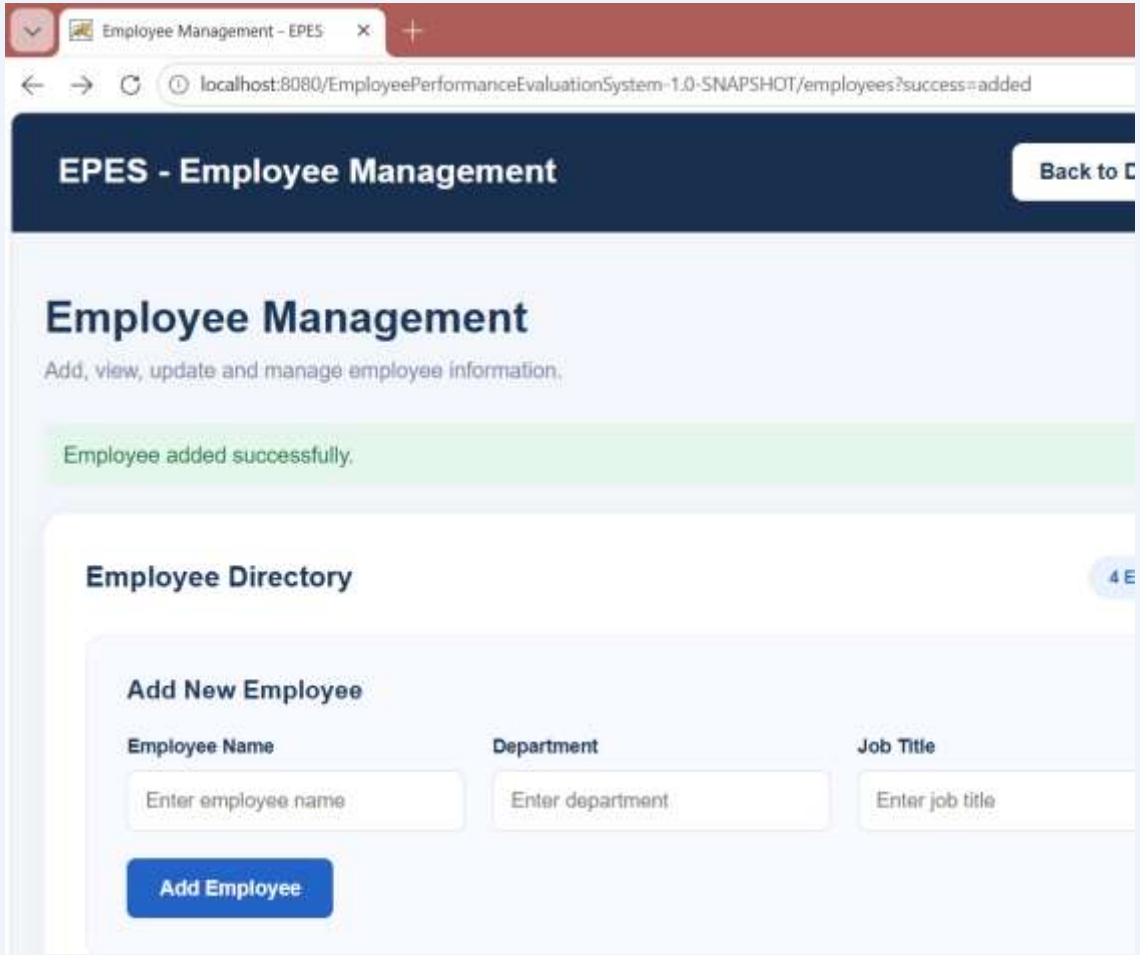

Manager dashboard and management modules

# 8. Application Screenshots — Employee & Directory

Employee self-service and employee administration



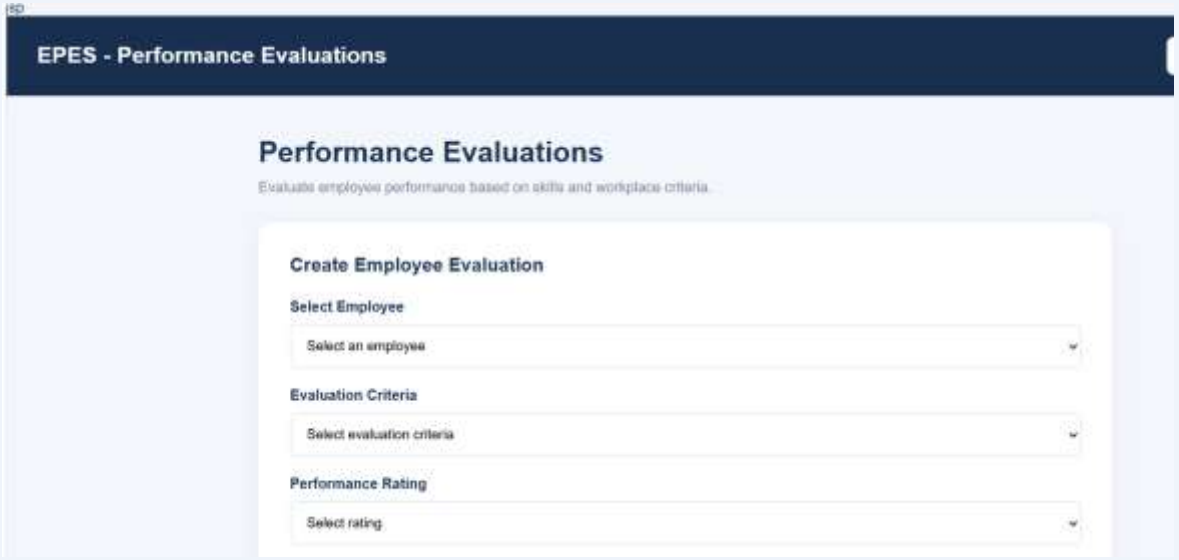

Employee dashboard: performance, goals, feedback and reports




Employee directory and add-employee form

# 9. Application Screenshots — Evaluation & Reports

Evidence of evaluation entry and calculated performance summary




Manager evaluation form with criteria and rating inputs




Performance report: evaluation count and average rating

# 10. Code Quality & Testing

Modular implementation and functional checks

## Code organization

- model: domain objects
- dao: database operations
- servlet: request handling
- util: DB, password, CSRF and HTML helpers

Prepared statements and try-with-resources support safer database access.

## Functional checks recorded

- Manager and Employee login
- Employee CRUD operations
- Manager-only route restrictions
- Employee-specific access
- Logout and protected-page behavior
- Invalid rating and goal status rejection
- CSRF token coverage on POST forms

Run the Maven build on the submission computer before final upload.

Note: The repository includes manual test utilities; these should not be described as a complete automated JUnit test suite.

# 11. Conclusion & Review 2 Outcome

EPES has progressed from the employee-management foundation shown in Review 1 to a role-based application covering key performance workflows.

### WORKING MODULES

Employee management
Performance evaluations
Goals and feedback
Performance summaries

### TECHNICAL FOUNDATION

Java 25 • Jakarta Servlets • JSP
MySQL 8 • JDBC • DAO
Tomcat 11 • Maven
Authentication and request security

### SUBMISSION MATERIAL

Public GitHub repository
README setup instructions
Database schema SQL
Application screenshots

Next improvements: password recovery, email notifications, downloadable PDF reports and analytics.

**GitHub: github.com/ShivamUpadhyay303/EmployeePerformanceEvaluationSystem**